

Priya Aryav

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Summary

- Software Engineer transitioning into ML Research Engineering, focused on LLM systems across pre-training, post-training, and inference optimization. Builds transformer training pipelines and CUDA-optimized kernels from scratch, backed by 5+ years of backend experience across microservices and web applications.

Projects

- Building llmdotc, an LLM training engine implemented from scratch in raw C and CUDA, with custom CUDA kernels for core transformer operations, focused on maximizing GPU compute throughput for pre-training workloads.
- tinyGPT: Implemented four decoder-only transformer architectures from scratch in PyTorch (bigram, GPT, GPT-2, Qwen2.5 with grouped query attention), with distributed training pipelines using DDP, mixed precision, gradient clipping, and cosine LR scheduling, benchmarking model FLOPs utilization across multi-GPU runs. Implemented key-value caching for efficient autoregressive inference and generation.
- Implemented tinybpe, a byte pair encoding tokenizer from scratch in Python, supporting basic and regex-based tokenization, special token registration, and model save and load functionality, validated by replicating OpenAI's GPT-4 tokenizer output exactly against production tiktoken results, with test coverage using pytest.
- Built opengrad, a minimal reverse-mode automatic differentiation engine supporting dynamic computation graphs and backpropagation, enabling training of small neural networks from first principles.
- Authored technical write-ups on transformer architecture, attention mechanisms, and tokenization, published on [Bear Blog](#).

Skills

Languages: **Python, C, C++, Java**
Frameworks: **PyTorch, CUDA**
ML Systems: **Pre-training, Transformers Architecture, Inference optimization**
Systems: **Linux, Distributed Systems, Performance Profiling**

Work Experience

Bengaluru, India

Lowe's - Software Engineer

Dec 2025 - Present

- Designed and implemented a distributed Java/Python-based data integration platform replacing Informatica, eliminating proprietary dependencies and saving \$4M annually through scalable architecture and performance optimization.

IBM - Software Engineer

June 2024 - Dec 2025

- Built real-time distributed data pipelines processing 1B+ records, reducing latency by 90%, and led adoption of Spock and Groovy testing frameworks across microservices, improving reliability by 30% and cutting production defects by 25%.

TCS - Software Engineer

Oct 2020 - June 2024

- Designed and scaled backend microservices and search systems (Spring Boot, Elasticsearch), handling 10K+ daily API calls while reducing response time by up to 50% and improving conversion by 14% through query optimization and caching.
- Led migration of six Tibco services to Spring Boot microservices, improving system efficiency 4x, increasing Salesforce Bulk API throughput by 50%, and implementing monitoring/alerting that reduced downtime by 80%.

Education

Bengal College of Engineering and Technology, Durgapur

Aug 2016 - July 2020

B.Tech in Computer Science and Engineering